

Swinburne University of Technology
School of Science, Computing, and Emerging Technologies

SWE40006 - Software Deployment and Evolution
Deployment Portfolio Task 2

Student name: Ngoc Van Nhi Do
Student ID: 105551765
Submission date: [DO NOT FORGETTTTTTTTTTTTTTTTTTTTTTTTTTT]

Deployment Portfolio Task 2

1 Declared Target Level

All four subtasks 2.1, 2.2, 2.3, and 2.4 were attempted.

Table 1 below contain relevant links as proof of successful task execution.

Table 1: Publicly accessible URLs for live grading verification

Source	URL
Task 2.1	https://swe40006-task-2-1-atbmg9a7daapdnec.australiaeast-01.azurewebsites.net
Task 2.2	https://swe40006-task-2-2-f9emhph2gddzhker.australiaeast-01.azurewebsites.net/

2 Execution Evidence

2.1 Task 2.1

For Task 2.1, I deployed a simple Node.js web application to Azure App Services, using Visual Studio Code and the Azure App Services extension.

First, I created the Node.js app consisting of the files `index.html`, `style.css`, `package.json` and `server.js`.



Figure 1: Structure of my simple Node.js application.

Then I ran my web application to test that it works correctly.

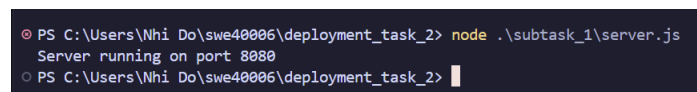


Figure 2: Running `npm start` on my Node.js application.

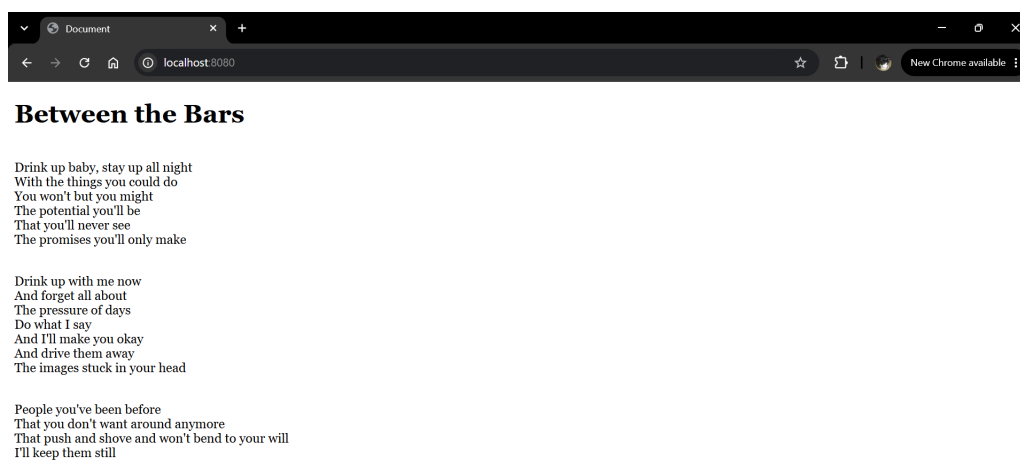


Figure 3: The web application runs locally on port 8080.

I then installed the Azure App Service on VS Code. The screenshot below shows the Azure panel once the extension has been successfully installed and I have logged in with my Azure account.

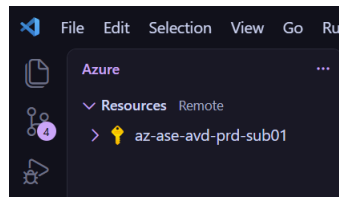


Figure 4: The Azure side panel on VS Code.

Next, I subscribed to an Azure for Students subscription on the Azure portal (portal.azure.com).

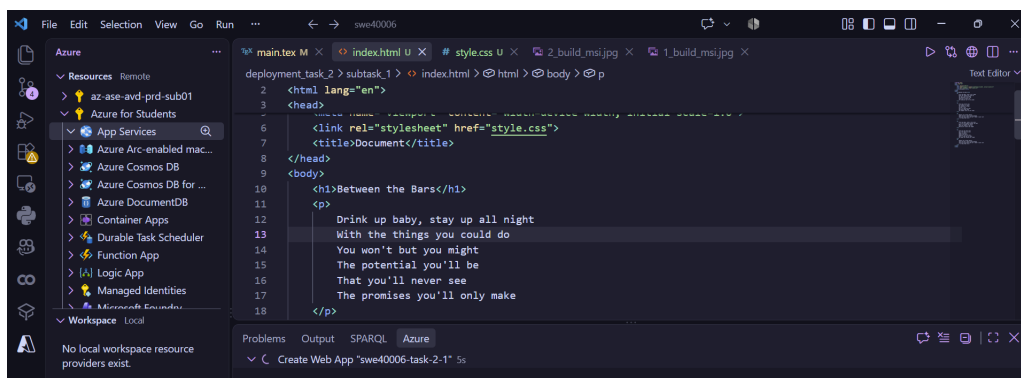


Figure 5: The Azure panel updated with my new Azure for Students subscription.

I then created a new Azure Web App service using the "Create Web App" command. I specified the region where I wanted my web app to be hosted which was Australia East, the application name `swe40006-task-2-1`, and the technology stack which was Node LTS 24.

As seen below, initial attempts to create the web app failed as my subscription was not registered with the `Microsoft.OperationalInsights` namespace. This was resolved by registering with the namespace on the Azure Portal.

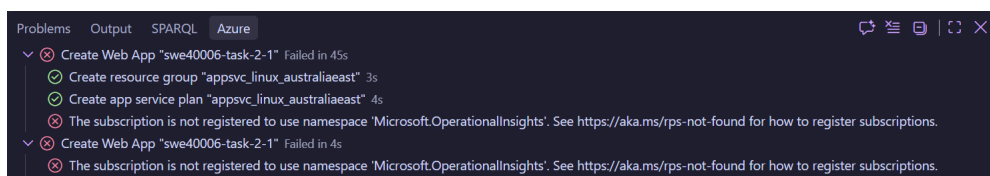


Figure 6: Failed attempts at creating a new Azure App Service instance.

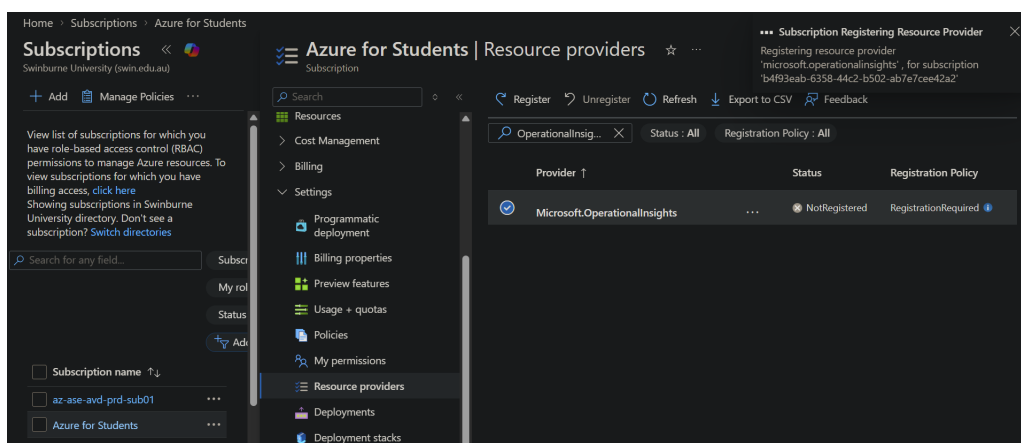


Figure 7: Registering with the `Microsoft.OperationalInsights` namespace on the Azure Portal.

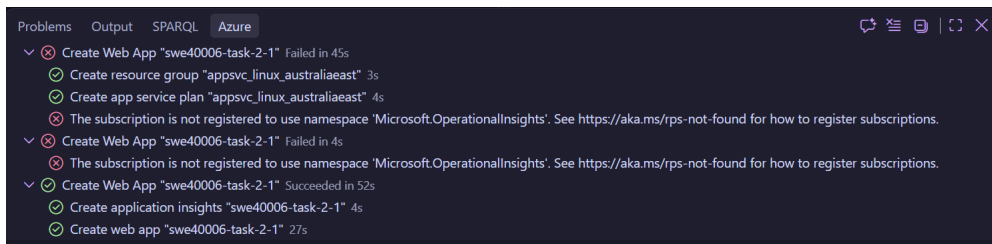


Figure 8: A new Azure App Service instance is successfully created.

The next step was to deploy my web application to the App Service instance.

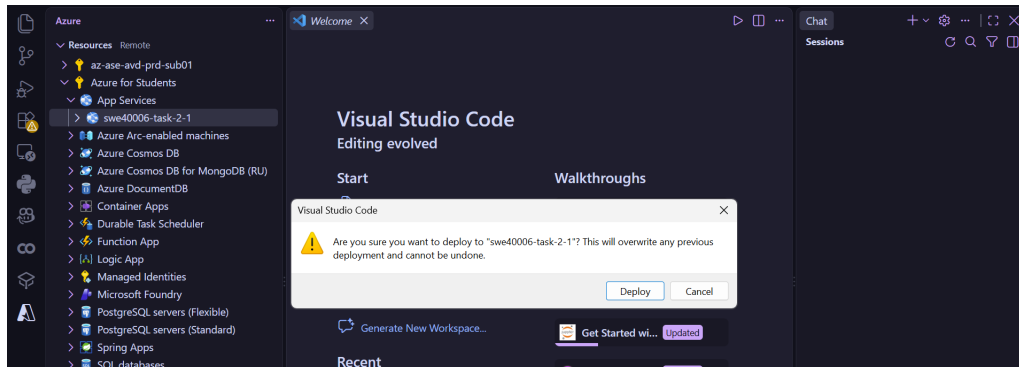


Figure 9: Deploying my web application to swe40006-task-2-1.

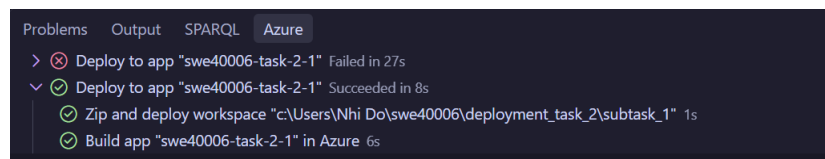


Figure 10: Deployment successful.

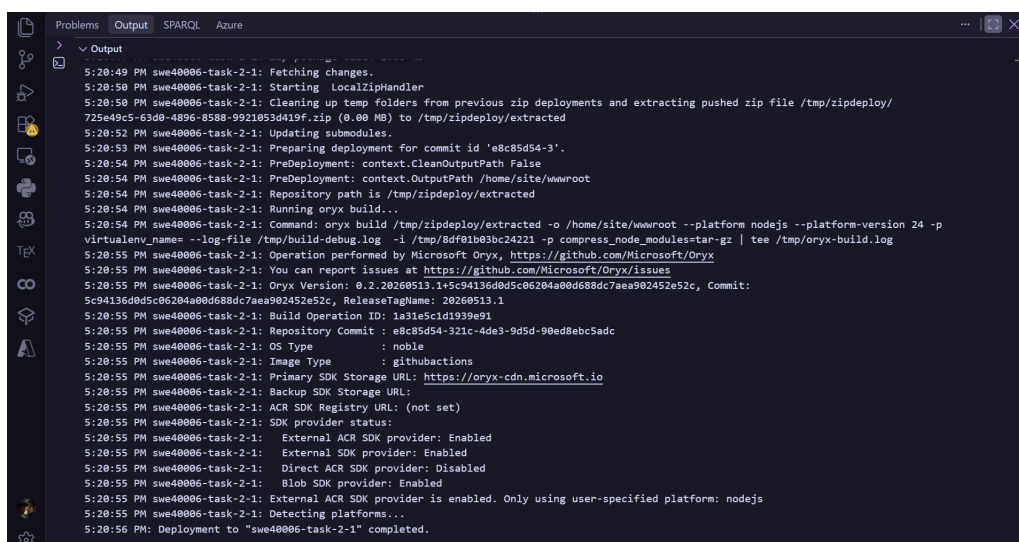


Figure 11: Build output.

The application is live and publicly accessible at <https://swe40006-task-2-1-atbmg9a7daapdnec.australiaeast-01.azurewebsites.net>.

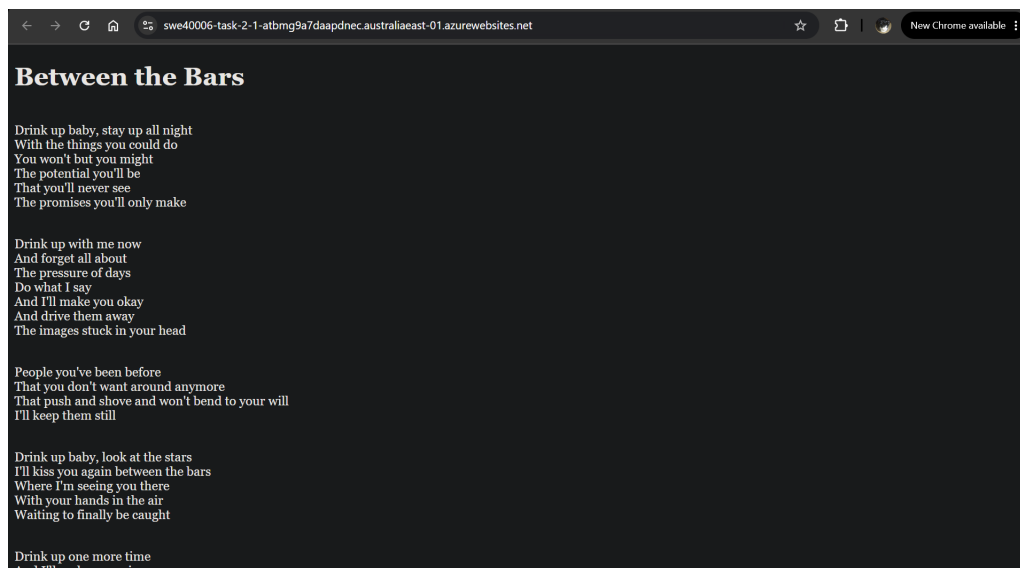


Figure 12: The application is deployed and running live.

2.2 Task 2.2

For task 2.2, I deployed an ASP.NET Razor Pages application with Azure App Services, also using Visual Studio Code and the Azure App Service extension.

First, I created a new Razor Pages project using the .NET CLI. Once the project has been created, I ran `dotnet publish` to publish the application.

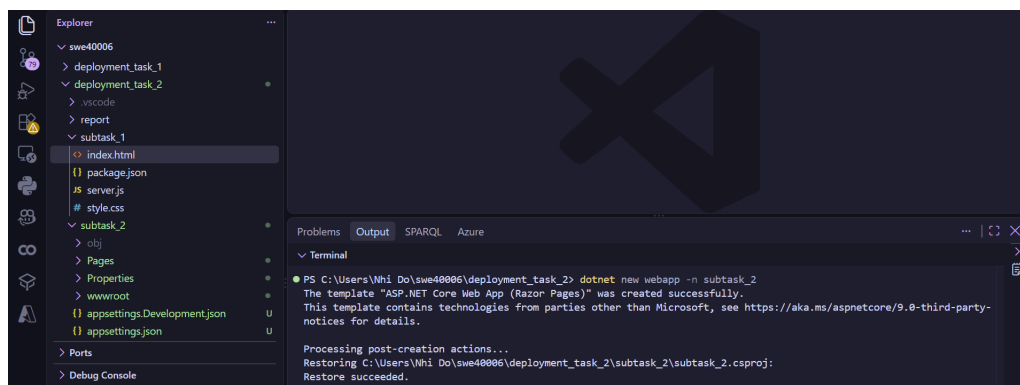


Figure 13: Creating a new Razor Pages project via the CLI.

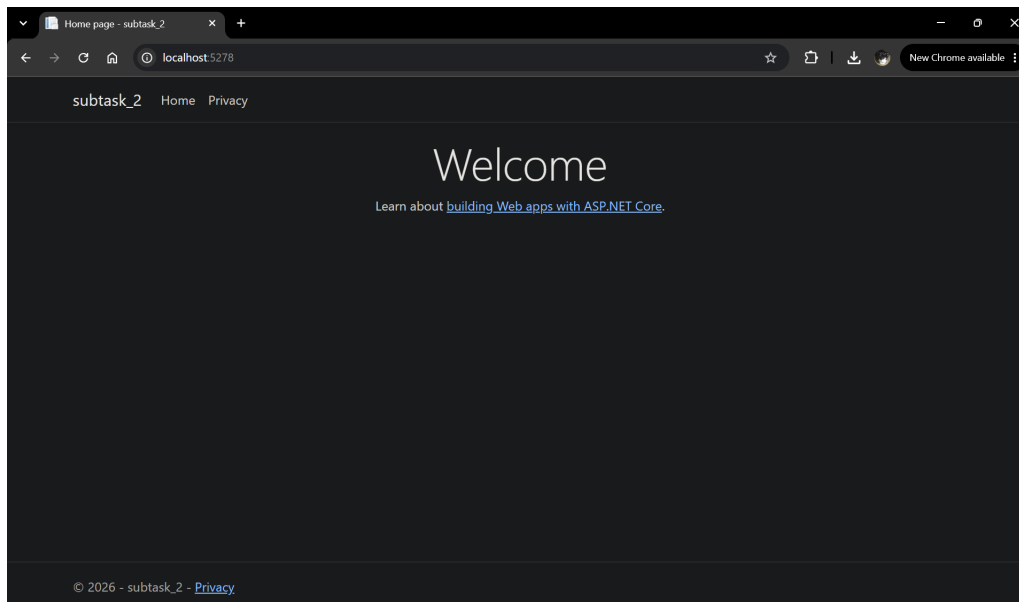


Figure 14: The application runs locally on port 5278.

Similar to Task 2.1, I created a new Web App Service instance using the Azure App Service extension. The new instance is named **swe40006-task-2-2**, hosted in the Australia East region, and using the .NET 8.0 runtime.

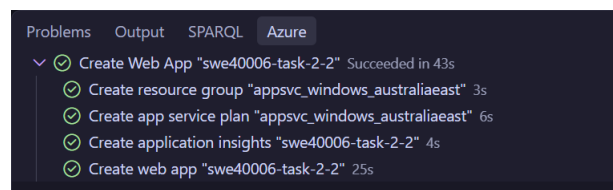


Figure 15: A new Web App Service instance is successfully created.

The application is then deployed into this new instance. It is hosted at <https://swe40006-task-2-2-f9emhph2gddzhker.australiaeast-01.azurewebsites.net/>.

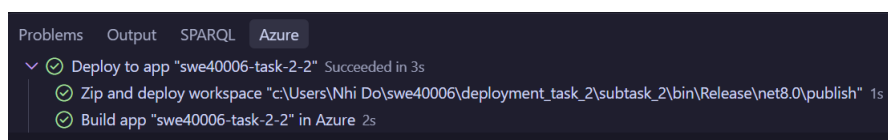
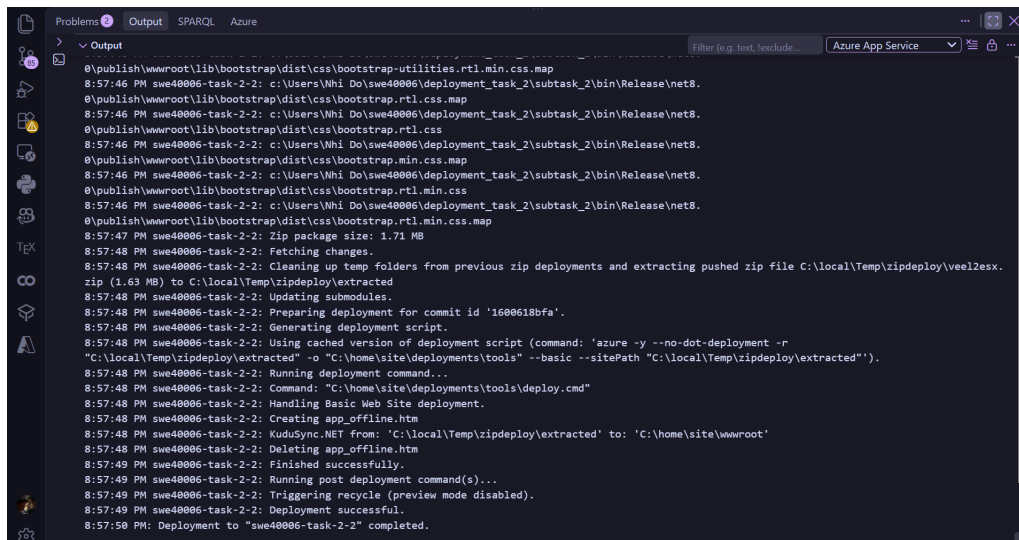


Figure 16: Successful deployment.



```

0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap-utilities.rtl.min.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.css
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.min.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.min.css
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.min.css.map
8:57:47 PM swe40006-task-2-2: Zip package size: 1.71 MB
8:57:48 PM swe40006-task-2-2: Fetching changes.
8:57:48 PM swe40006-task-2-2: Cleaning up temp folders from previous zip deployments and extracting pushed zip file C:\local\Temp\zipdeploy\veel2esx.
zip (1.63 MB) to C:\local\Temp\zipdeploy\extracted
8:57:48 PM swe40006-task-2-2: Updating submodules.
8:57:48 PM swe40006-task-2-2: Preparing deployment for commit id '1600618bfa'.
8:57:48 PM swe40006-task-2-2: Generating deployment script.
8:57:48 PM swe40006-task-2-2: Using cached version of deployment script (command: 'azure -y --no-dot-deployment -r
"C:\local\Temp\zipdeploy\extracted" -o "C:\home\site\deployments\tools" --basic --sitePath "C:\local\Temp\zipdeploy\extracted"').
8:57:48 PM swe40006-task-2-2: Running deployment command...
8:57:48 PM swe40006-task-2-2: Command: "C:\home\site\deployments\tools\deploy.cmd"
8:57:48 PM swe40006-task-2-2: Handling Basic Web Site deployment.
8:57:48 PM swe40006-task-2-2: Creating app_offline.htm
8:57:48 PM swe40006-task-2-2: KuduSync.NET from: 'C:\local\Temp\zipdeploy\extracted' to: 'C:\home\site\wwwroot'
8:57:48 PM swe40006-task-2-2: Deleting app_offline.htm
8:57:49 PM swe40006-task-2-2: Finished successfully.
8:57:49 PM swe40006-task-2-2: Running post deployment command(s)...
8:57:49 PM swe40006-task-2-2: Triggering recycle (preview mode disabled).
8:57:49 PM swe40006-task-2-2: Deployment successful.
8:57:50 PM: Deployment to "swe40006-task-2-2" completed.

```

Figure 17: Logs showing that deployment was successful.

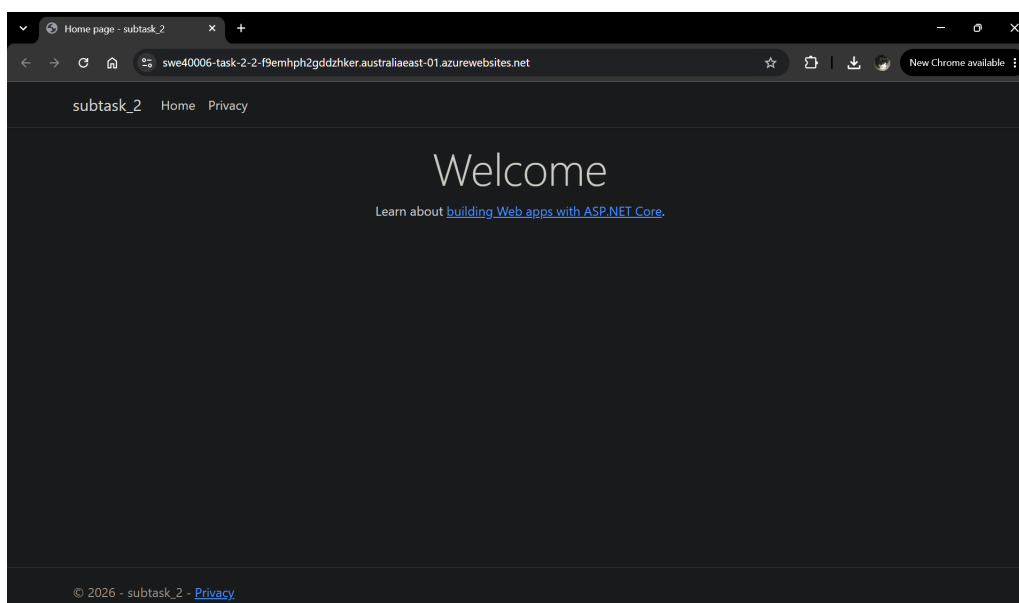


Figure 18: The web application up and running live.

Even though this App Service instance is running on a free tier, it is still best practice to manage cloud resources responsibly to avoid unintended charges. To demonstrate this, I stopped the instance using the Azure Portal.

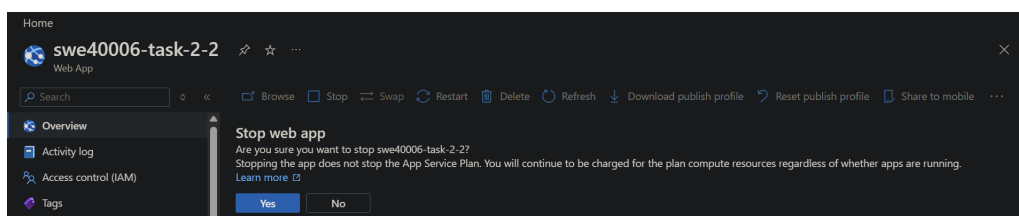


Figure 19: Stopping the instance.

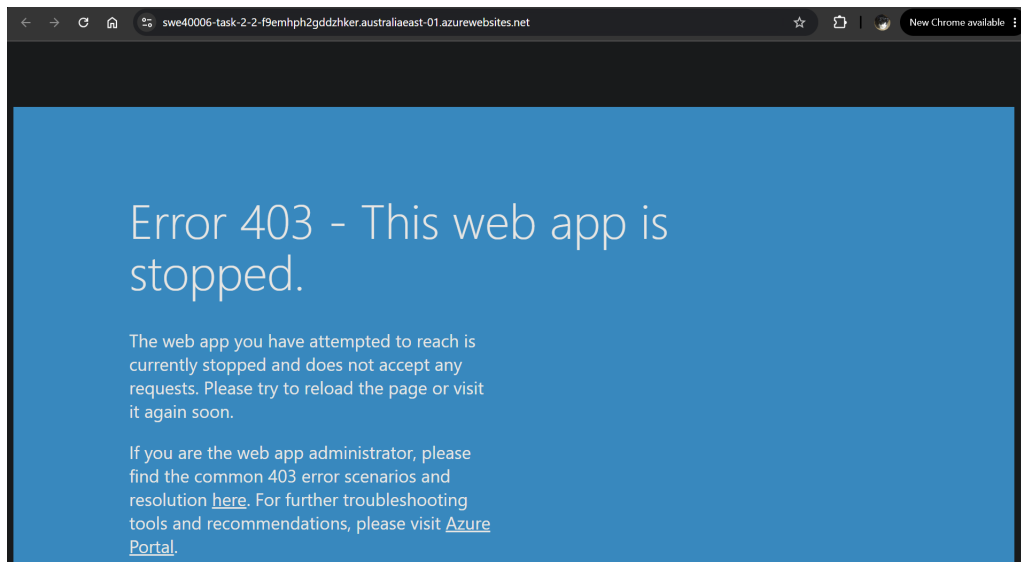


Figure 20: The web app has been stopped.